

Postprocessing Data Input Format

This document describes the data format required for QKD postprocessing. The system expects two types of data streams, typically provided through FIFOs (First-In, First-Out) or files on the filesystem: Angle data and Click Result data.

Angle Data

Angle data is used to determine the basis choices for quantum state preparation and measurement. Both Alice's and Bob's systems must provide angle data in the same format.

- **Source:** The angle data should be made available as a stream of bytes, typically from a character device file (e.g., `/dev/xdma0_c2h_1`) acting as a FIFO.
- **Grouping:** The postprocessing software reads data in batches of 32 angles.
- **Encoding:** Each batch of 32 angles is encoded into a 16-byte array (`[u8; 16]`).
- **Hardware Writes:** Hardware can write data to the FIFO in any chunk size. The FIFO buffers the byte stream, so hardware writes are not required to be aligned with the 16-byte read chunks.

Bit-level Format

Within the 16-byte array, each byte packs two angle values. The angles are 2-bit values.

For each byte B in the angle data array:

- The angle for event $2i$ is encoded in the lower two bits of B : $\alpha_{2i} = (B \ \& \ 0b0000_0011)$.
- The angle for event $2i + 1$ is encoded in bits 4 and 5 of B : $\alpha_{2i+1} = (B \ \& \ 0b0011_0000) \gg 4$.

Where i is the index of the byte in the 16-byte array.

The bits of each byte are structured as follows:

- Bits 0-1: First angle value (event $2i$).
- Bits 2-3: Unused.
- Bits 4-5: Second angle value (event $2i + 1$).
- Bits 6-7: Unused.

Physical Interpretation of Angles

The 2-bit angle values provided by both Alice and Bob correspond to local phase rotations applied to a quantum state. In this system, the initial state is considered to be $|+\rangle$.

Each party (Alice and Bob) applies a local rotation to the state. Let Alice's rotation angle be α_A and Bob's be α_B . These are selected from a set of four angles, typically $\{0, \pi/2, \pi, 3\pi/2\}$, corresponding to the 2-bit integer values 0, 1, 2, 3.

Alice prepares a state by applying her rotation:

$$|\psi_A\rangle = \exp(i\alpha_A\sigma_y) |+\rangle \quad (1)$$

This state is then transmitted to Bob, who applies his own local rotation. The final state, before measurement, is determined by the sum of their angles:

$$|\psi_{\text{final}}\rangle = \exp(i(\alpha_A + \alpha_B)\sigma_y) |+\rangle \quad (2)$$

Here, σ_y is the Pauli-Y operator. The probability of a detection event (a "click") at Bob's end depends on this final state.

This choice of angles is analogous to basis selection in other QKD protocols like BB84.

- When $\alpha_A + \alpha_B$ is 0 or π (modulo 2π), Alice and Bob are considered to have chosen the **same basis**. In this case, the measurement outcome is deterministic.
- When $\alpha_A + \alpha_B$ is $\pi/2$ or $3\pi/2$ (modulo 2π), they are considered to have chosen **different bases**, leading to a random measurement outcome.

This mechanism allows for post-processing to sift the key, keeping only the results where the same basis was used, and to calculate the postprocessing from a sample of the results from different bases.

Click Result Data

The click result data stream indicates the outcome of measurements. This is typically only relevant on Bob's side.

- **Source:** The click results are expected from a FIFO, which can be a standard Linux named pipe or a device file.
- **Format:** The data is a stream of bytes. Each byte represents a single click result.
- **Value:**
 - A byte value of 0 represents one outcome (e.g., no click).
 - A byte value of 1 represents the other outcome (e.g., click).
- **Grouping:** The results are read in batches of 32, where each batch corresponds to a batch of 32 angles from both Alice and Bob. The system will read a [u8; 32] array for each batch.
- **Hardware Writes:** As with angle data, hardware can write click results to the FIFO in any chunk size.

Data Synchronization

It is crucial that all data streams (Alice's angles, Bob's angles, and click results) are perfectly synchronized. The software assumes a one-to-one correspondence: the Nth angle value from Alice, the Nth angle value from Bob, and the Nth click result must all correspond to the same single physical measurement event. Any misalignment will lead to incorrect postprocessing results.